

MAYANK TYAGI

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[LinkedIn](#) · [Codeforces](#) · [HackerRank](#)

SUMMARY

Software Engineer with hands-on experience in embedded systems and real-time communication, currently working on BLE stack integration and system optimization at Espressif. Strong problem-solving background with **AIR 85 at ICPC Asia West Regionals**. Skilled in building efficient, low-level systems with a focus on performance, memory optimization, and scalable software design.

EDUCATION

Bachelor of Engineering, Army Institute of Technology, Pune

Expected 2026

Branch: Electronics and Telecommunication Engineering

SGPA: 8.3

WORK EXPERIENCE

Software Engineer, Espressif Systems

Jan 2026 – Present

- Developed BLE stack features and bug fixes in C, gaining hands-on experience with GAP/GATT procedures, event-driven architecture, and real-time device communication.
- Replaced semaphore-based synchronization with FreeRTOS task notifications in user features, reducing RAM usage and improving system efficiency in constrained environments.
- Designed and implemented integration between BLE and ESP-Insights, enabling real-time log streaming from devices to the ESP-Insights web dashboard.
- Investigated and resolved multiple customer-reported BLE HID interoperability issues through direct collaboration with customers, identifying root causes and delivering fixes for battery level service behavior, HID profile interactions, and client-side notification/event delivery failures, improving product stability and customer satisfaction.
- Performed performance analysis on the NimBLE GATT throughput application and optimized BLE parameters, improving throughput by 200+ kbps.

Problem Setter, GeeksforGeeks

April 2024 - July 2024

- Created and curated original **competitive programming** problems focused on core data structure concepts.
- Contributed to **national-level coding contests** on GeeksforGeeks, attempted by over **5000 participants**.

PROJECTS

Browser History Design

[GitHub](#)

Tech stack: (Doubly Linked List, OOP)

Designed a browser history system with doubly linked list to support back/forward navigation and permanent history log. Applied OOP principles like encapsulation (private Node), abstraction (history operations), and modular class design.

Tab Optimizer

[GitHub](#)

Tech stack: JavaScript, Chrome Extension APIs, C++

Developed a Chrome extension to optimize browser tab resource usage by monitoring memory and CPU consumption. Implemented automatic suspension of inactive tabs to reduce system overhead and improve performance. Integrated a lightweight C++ module for enhanced system-level resource tracking.

TECHNICAL ACHIEVEMENTS

- **Achieved All India Rank 85** at the **ICPC Asia West Regionals, Chennai Site (2024–2025)**. [ICPC Profile](#)
- **Solved 1000+ problems** across Codeforces, CodeChef, LeetCode, and HackerRank. [Profile](#)
- **Secured Rating of Expert (1655)** on Codeforces, demonstrating strong problem-solving skills. [Profile](#)
- **Secured global rank 865** in Round 935 (Div. 3) and **975** in Round 993 (Div. 4).
- **4-Star (1800 rated)** coder on CodeChef with consistent performance in rated contests. [Profile](#)
- **Top 100 finalists in Codeft (National Level)**: Ranked twice in National-level coding competition. [Certificate](#)
- **Secured 3rd position CRACK 5.2**, a college-level competitive coding competition.
- **GameJam Hackathon**: Secured **3rd position** as part of a team; contributed to UI design and game development.

SKILLS

Technical Skills: Competitive Programming, Data Structures and Algorithms, Operating Systems, Object-Oriented Programming, Computer Networks, DBMS.

Programming Languages: C++, Python, Java, C

Programming Tools: Git, GitHub, MongoDB, Oracle SQL, Streamlit, TMDb API

Soft Skills: Teamwork, Adaptability, Leadership, Critical Thinking, Planning and Organizing